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DHT Sensor and OLED with MKR Connector Carrier

Learn how to display temperature and Humidity on an OLED display with the MKR Connector Carrier.

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This basic example teaches you how to create a circuit that uses two Grove modules and requires no soldering. The MKR board you choose may be anyone of the MKR family because the connection is managed through the MKR Connector Carrier.

Hardware Required

[Arduino MKR family board](#)

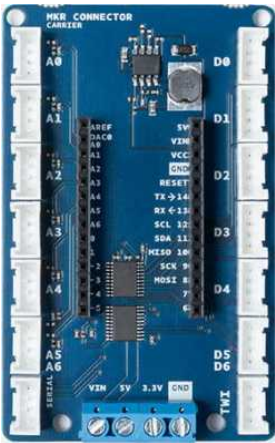
[Arduino MKR Connector Carrier](#)

[Grove DHT Module](#)

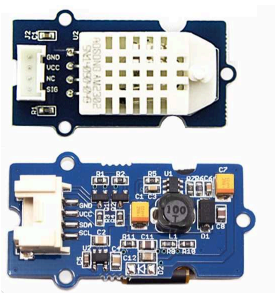
[Grove OLED 0.96"](#)

Circuit

The connection to the Carrier board requires two standard Grove cables. The DHT humidity and temperature sensor goes to the D0 connector. The OLED screen is connected to the TWI connector.



The MKR Connector Carrier.



Grove DHT sensor module.

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We did not put a MKR board on the Carrier, but it is required to get the circuit to work, as specified in the bill of materials.

The DHT module uses a specific pin to communicate with the MKR board and it is mapped on D0. This happens because the Grove standard for digital connections follows this rule:

PinFuncrionNotespin1DnPrimary digital i/opin2Dn+1Secondary digital i/opin3VCCPower to module 5V/3.3Vpin4GNDGround

and the module sends SIG on pin 1 that is mapped on the primary digital I/O.

In the picture, the OLED module is shown from component side to let you see the Grove connector. It

MKR Connector Carrier that follows this pin mapping:

PinFunctionNotespin1SCLI2C
Clockpin2SDAI2C
Datapin3VCCPower to module
5V/3.3Vpin4GNDGround

Code

To drive the modules you need to load four separate libraries:

```
1 #include <DHT.h>
2 #include <DHT_U.h>
3 #include <Wire.h>
4 #include <SeeedOLED.h>
```

The DHT module is mapped on D0 when the object dht is instantiated:

```
1 DHT dht(0, DHT22);
```

The rest of the code is straightforward and keeps reading the `hum` and `temp` values to be printed on the OLED screen.

Here is the complete sketch:

```
1 #include <DHT.h>
2 #include <DHT_U.h>
3 #include <Wire.h>
4 #include <SeeedOLED.h>
5
6 DHT dht(0, DHT22);
7
8 void setup() {
9
10   Wire.begin();
11
12   SeeedOled.init();
13
14   SeeedOled.clearDisp
15
16   SeeedOled.setNormal
17
18   SeeedOled.setPageMo
19 }
20
21 void loop() {
22
23   float temp, hum;
24
25   //Read temperature
26
27   do {
28
29     hum = dht.readHum
```

Suggest changes

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